

PROJECT: PREDICTIVE MAINTENANCE FOR INDUSTRIAL EQUIPMENT

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Context

In industrial settings, equipment failures can lead to significant production losses and increased maintenance expenses. Traditional maintenance strategies, such as reactive and preventive maintenance, are often inefficient and costly. Predictive maintenance leverages data analytics and machine learning to predict when equipment is likely to fail, allowing for timely interventions.

Objective

The goal of this project is to develop a predictive maintenance system that can forecast equipment failures before they occur, thereby minimizing downtime and maintenance costs.

Dataset Description

The dataset consists of 10 000 data points stored as rows with 14 features in columns:

- UID: unique identifier ranging from 1 to 10000.
- Product ID: consisting of a letter L, M, or H for low (60% of all products), medium (30%) and high (10%) as product quality variants and a variant-specific serial number.
- Air temperature [K]: generated using a random walk process later normalized to a standard deviation of 2 K around 300 K.
- Process temperature [K]: generated using a random walk process normalized to a standard deviation of 1 K, added to the air temperature plus 10 K.
- Rotational speed [rpm]: calculated from a power of 2860 W, overlaid with a normally distributed noise.
- Torque [Nm]: torque values are normally distributed around 40 Nm with a standard deviation of 10 Nm and no negative values.
- Tool wear [min]: The quality variants H/M/L add 5/3/2 minutes of tool wear to the used tool in the process.
- Machine failure: label that indicates, whether the machine has failed in this particular data point for any of the following failure modes are true. The machine failure consists of five independent failure modes:
 - tool wear failure (TWF): the tool will be replaced or fail at a randomly selected tool wear time between 200 - 240 mins.
 - heat dissipation failure (HDF): heat dissipation causes a process failure, if the difference between air- and process temperature is below 8.6 K and the tools rotational speed is below 1380 rpm.
 - power failure (PWF): the product of torque and rotational speed (in rad/s) equals the power required for the process. If this power is below 3500 W or above 9000 W, the process fails.

- overstrain failure (OSF): if the product of tool wear and torque exceeds 11,000 minNm for the L product variant (12,000 M, 13,000 H), the process fails due to overstrain.
- random failures (RNF): each process has a chance of 0,1 % to fail regardless of its process parameters. If at least one of the above failure modes is true, the process fails and the 'machine failure' label is set to 1. It is therefore not transparent to the machine learning method, which of the failure modes has caused the process to fail.

Proposed Methodology

1. Preprocessing:

- Handle missing values, null values and duplicate values.
- Removing unimportant columns.

2. Exploratory Data Analysis (EDA):

- Understand the distribution of 'Product Type', 'Failure Type', 'Target' variables.
- Visualize the distribution of Failure Types.
- Summary statistics for the data.
- Check the correlation among the variables.

3. Modelling:

Test different machine learning algorithms, such as:

- Decision Tree
- Random Forest
- K-Nearest Neighbours
- Logistic Regression
- Support Vector Machine

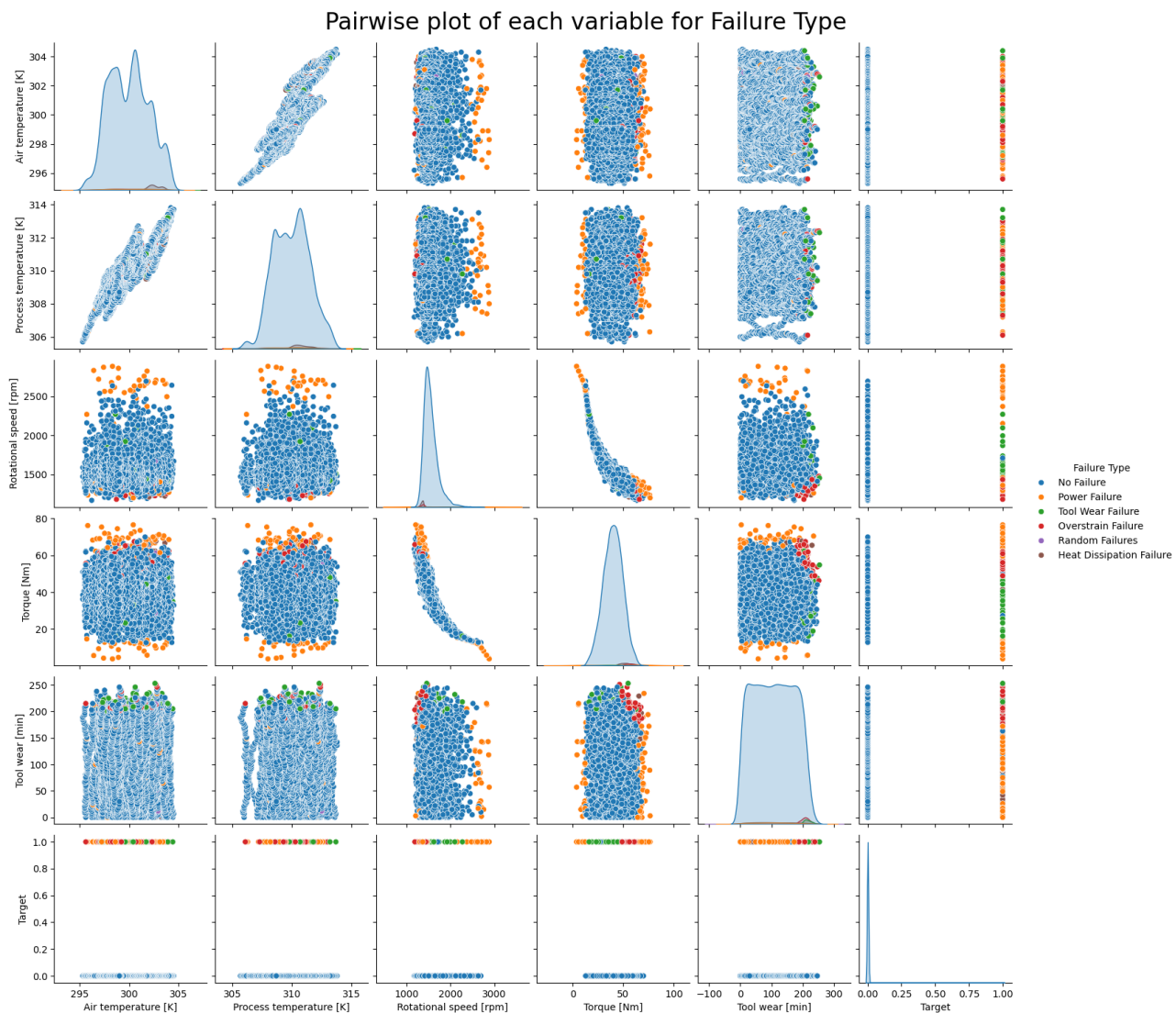
Evaluation

- Use appropriate parameters like model accuracy, precision, recall, and F1-score to evaluate performance.
- Analyse the confusion matrix to identify error patterns.

Importing and data preprocessing

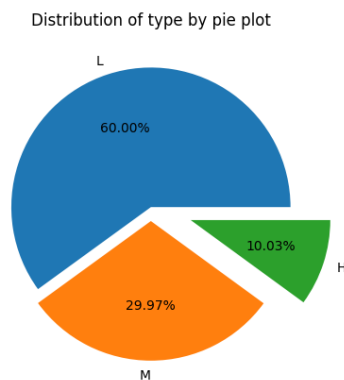
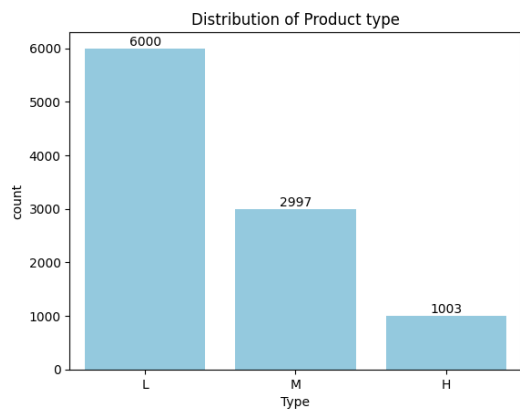
I chose Kaggle dataset for the task, after choosing the most suited dataset the preparation phase begins, the preparation of the dataset includes selecting the wanted attributes or variables, cleaning it by excluding Null rows, deleting duplicated variables, treating outlier if necessary, in addition to transforming data types to the wanted type, data merging can be performed as well where two or more attributes get merged. All those alterations lead to the wanted result which is to make the data ready to be modeled.

There were no null values nor duplicate values present in the dataset. Unimportant columns were deleted. Data type of float and integer was selected for further analysis. 'Failure type' and 'target' variables later merged as it was our targeted variables. There are total 339 (3.39%) failures and 9661 (96.61%) no failures. Also removed data with contradictory target and failure type variables, we are remain with 9973 valid records. Following is the visual presentation for each variable according to failure type.

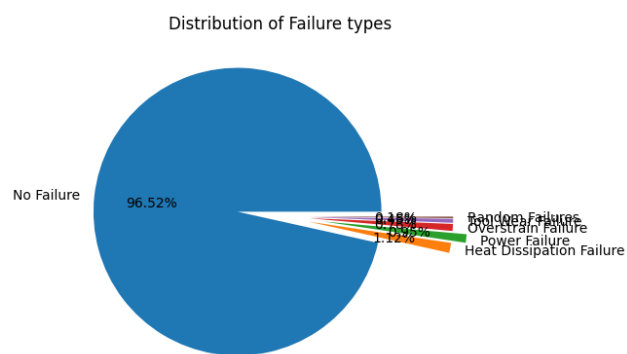
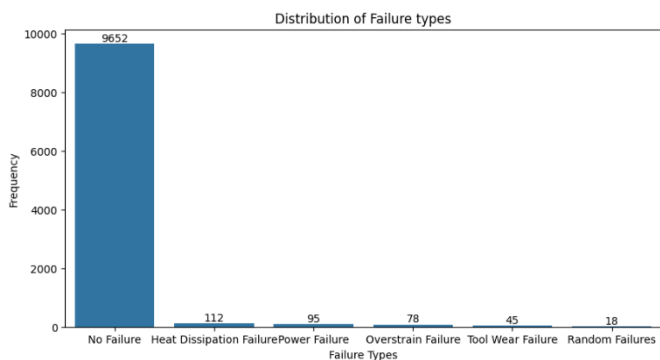


Exploratory Data Analysis (EDA):

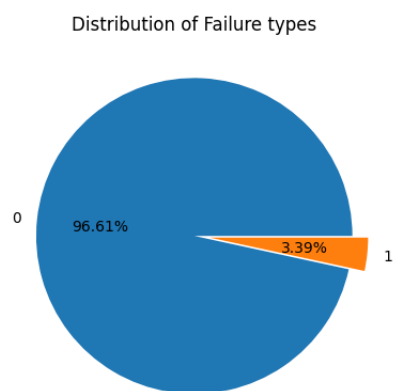
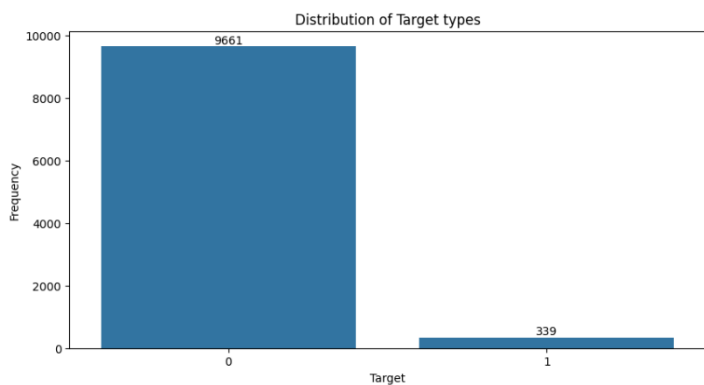
1) There are 6000 (60.00%) low quality products, 2997 (29.97%) medium quality products and 1003 (10.03%) low quality products



2) There are 9652 no failures, 112 heat dissipation failure, 95 power failures, 78 overstrain failures, 45 tool wear failures, 18 random failures.



3) There are total 339 (3.39%) failures and 9661 (96.61%) no failures



The basic statistics for the data is obtained which include mean, standard deviation, quartiles, minimum and maximum for each variable for fraud and valid transactions. Following inferences can be drawn with the help of the statistics:

	count	mean	std	min	25% \
Air temperature [K]	10000.0	300.00493	2.000259	295.3	298.3
Process temperature [K]	10000.0	310.00556	1.483734	305.7	308.8
Rotational speed [rpm]	10000.0	1538.77610	179.284096	1168.0	1423.0
Torque [Nm]	10000.0	39.98691	9.968934	3.8	33.2
Tool wear [min]	10000.0	107.95100	63.654147	0.0	53.0
Target	10000.0	0.03390	0.180981	0.0	0.0

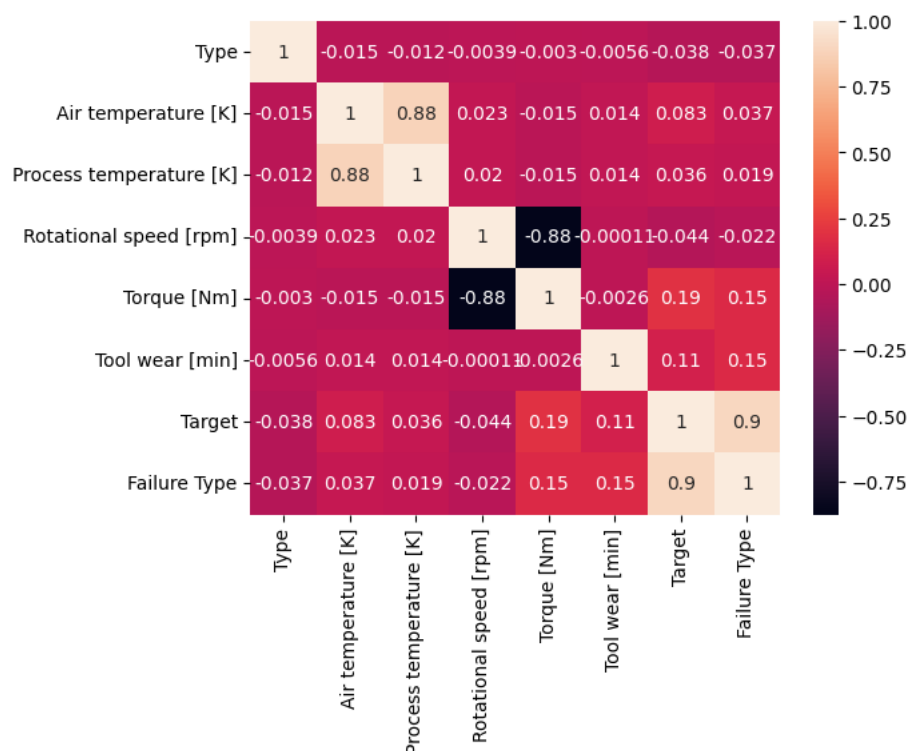
	50%	75%	max
Air temperature [K]	300.1	301.5	304.5
Process temperature [K]	310.1	311.1	313.8
Rotational speed [rpm]	1503.0	1612.0	2886.0
Torque [Nm]	40.1	46.8	76.6
Tool wear [min]	108.0	162.0	253.0
Target	0.0	0.0	1.0

Encode the variable data into integer data for better evaluation.

For Type of product quality: 'L':1,'M':2,'H':3
For Type of failure: 'No Failure': 0,'Heat Dissipation Failure': 1,'Power Failure': 2, 'Overstrain Failure': 3, 'Tool Wear Failure': 4

Checking the correlation among the variables.

Heatmap:- A heatmap is a graphical representation of data where values are depicted by colour. It helps visualize complex information in a way that's easy to comprehend. They're particularly useful for identifying patterns and trends within data, making them valuable. From the heat map it is evident that the correlation among the principle components is very low.



1. The variables Target and Failure Type are highly correlated, and since they are the target variables for our model, these variables will be combined into a single variable.
2. Process Temperature [K] and Air Temperature [K] are positively correlated.
3. Torque [Nm] and Rotational speed [rpm] are negatively correlated.

Merging the variables 'Target' and 'Failure Type' as 'TargetFailure' for better modelling.

'0-0': 0, '1-1': 1, '1-2': 2, '1-3': 3, '1-4': 4

Feature Selection

Take 'TargetFailure' as targeted variable and rest numerical variables as features.

train_test_split: A train-test split is a technique used in machine learning to evaluate the performance of a model. Here train 80% of data and test 20% of data.

Cross Validation:

Cross-validation is a technique used to assess the performance and generalizability of a machine learning model.

1. **Purpose:** To ensure that the model performs well on unseen data and is not overfitting to the training data.
2. **Process:**
 - **Split the Data:** Divide the dataset into multiple subsets or “folds” (commonly 5 or 10).
 - **Training and Validation:** Train the model on all but one fold and validate it on the remaining fold. Repeat this process for each fold.
 - **Average Results:** Calculate the average performance metrics (e.g., accuracy, precision) across all folds to get a more reliable estimate of the model’s performance.
3. **Benefits:** Provides a more robust evaluation by using all data points for both training and validation, reducing the risk of bias.

Result:

Cross-Validation Results (Accuracy): [1. 1. 1. 1. 1.]

Mean Accuracy: 1.0

This implies that the model has achieved perfect accuracy on all the validation sets used during the cross-validation process. This means that the model correctly predicted all the outcomes for each fold of the data.

Modeling:-

Five machine learning models were created in the modelling phase, Decision Tree, Random Forest Classifier, k Neighbours Classifier, Logistic Regression and Support Vector Classifier. A comparison of the results will be presented later in the paper to know which technique is most suited in the Predictive maintenance for industrial equipments. The dataset is sectioned into a ratio of 80:20, the training set will be the 80% and remaining set will be the testing set which is the 20%. The five models were created using libraries in python sklearn

1. Decision Tree:

A decision tree is a hierarchical model used in decision support that depicts decisions and their potential outcomes. It is a non-parametric, supervised learning algorithm that is useful for both classification and regression tasks. The tree structure in the decision model helps in drawing a conclusion for any problem which is more complex in nature. The decision rules are generally in the form of if-then-else statements. Decision trees are transparent, efficient, and flexible.

2. Random Forest:

Random Forest is a powerful ensemble learning technique used in machine learning for both classification and regression tasks.

Structure: It consists of multiple decision trees (hence the name “forest”) that work together to improve the model’s accuracy and robustness.

Working:

Training: Each tree in the forest is trained on a random subset of the data (bagging) and a random subset of features.

Prediction: For classification, each tree votes for a class, and the majority vote is taken as the final prediction. For regression, the average of all tree predictions is used.

3. k Neighbours Classifier

The k-Nearest Neighbours (k-NN) classifier is a simple, yet effective, machine learning algorithm used for classification tasks.

Concept: k-NN is a lazy learning algorithm that classifies a data point based on the majority class of its k nearest neighbours in the feature space.

Working:

Distance Calculation: For a given data point, calculate the distance (e.g., Euclidean distance) to all other points in the dataset.

Identify Neighbours: Select the k closest data points (neighbours).

Voting: For classification, the data point is assigned the class that is most common among its k nearest neighbours.

4. Logistic Regression

Logistic Regression model is statical model where evaluations are formed of the connection among dependent qualitative variable (binary or binomial logistic regression) or variable with three values or higher (multinomial logistic regression) and one independent explanatory variable or higher whether qualitative or quantitative.

5. Support Vector Classifier

SVC aims to find the optimal hyperplane that separates data points of different classes with the maximum margin. The margin is the distance between the hyperplane and the nearest data points from each class, known as support vectors.

Working:

Linear SVC: For linearly separable data, SVC finds a straight line (in 2D) or a hyperplane (in higher dimensions) that best separates the classes.

Non-linear SVC: For non-linearly separable data, SVC uses kernel functions (e.g., polynomial, radial basis function) to transform the data into a higher-dimensional space where a linear separation is possible.

Evaluation:

Model Accuracy: -

```
Decision Tree accuracy: 1.00
Random Forest accuracy: 1.00
K-Nearest Neighbors accuracy: 1.00
Logistic Regression accuracy: 1.00
Support Vector Machine accuracy: 1.00
```

Every model accuracy is 1 this implies that all your positive samples are classified as positive samples and none of the positive samples are classified incorrectly.

Classification Report: -

Evaluating Decision Tree...

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	1936
1	1.00	1.00	1.00	23
2	1.00	1.00	1.00	23
3	1.00	1.00	1.00	8
4	1.00	1.00	1.00	5
accuracy			1.00	1995
macro avg	1.00	1.00	1.00	1995
weighted avg	1.00	1.00	1.00	1995

Evaluating Random Forest...

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	1936
1	1.00	1.00	1.00	23
2	1.00	1.00	1.00	23
3	1.00	1.00	1.00	8
4	1.00	1.00	1.00	5

accuracy			1.00	1995
macro avg	1.00	1.00	1.00	1995
weighted avg	1.00	1.00	1.00	1995

Evaluating K-Nearest Neighbors...

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	1936
1	1.00	1.00	1.00	23
2	1.00	1.00	1.00	23
3	1.00	1.00	1.00	8
4	1.00	1.00	1.00	5

accuracy			1.00	1995
macro avg	1.00	1.00	1.00	1995
weighted avg	1.00	1.00	1.00	1995

Evaluating Logistic Regression...

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	1936
1	1.00	1.00	1.00	23
2	1.00	1.00	1.00	23
3	1.00	1.00	1.00	8
4	1.00	1.00	1.00	5

accuracy			1.00	1995
macro avg	1.00	1.00	1.00	1995
weighted avg	1.00	1.00	1.00	1995

Evaluating Support Vector Machine...

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	1936
1	1.00	1.00	1.00	23
2	1.00	1.00	1.00	23
3	1.00	1.00	1.00	8
4	1.00	1.00	1.00	5

accuracy			1.00	1995
macro avg	1.00	1.00	1.00	1995
weighted avg	1.00	1.00	1.00	1995

Classification report shows there is 100% precision for detecting failures for every type of failure. This implies that model has high accuracy in correctly identifying positive outcomes.

Confusion Matrix: -

Confusion Matrix for Decision Tree:

```
[[1936    0    0    0    0]
 [    0   23    0    0    0]
 [    0    0   23    0    0]
 [    0    0    0    8    0]
 [    0    0    0    0    5]]
```

Confusion Matrix for Random Forest:

```
[[1936    0    0    0    0]
 [    0   23    0    0    0]
 [    0    0   23    0    0]
 [    0    0    0    8    0]
 [    0    0    0    0    5]]
```

Confusion Matrix for K-Nearest Neighbors:

```
[[1936    0    0    0    0]
 [    0   23    0    0    0]
 [    0    0   23    0    0]
 [    0    0    0    8    0]
 [    0    0    0    0    5]]
```

Confusion Matrix for Logistic Regression:

```
[[1936    0    0    0    0]
 [    0   23    0    0    0]
 [    0    0   23    0    0]
 [    0    0    0    8    0]
 [    0    0    0    0    5]]
```

Confusion Matrix for Support Vector Machine:

```
[[1936    0    0    0    0]
 [    0   23    0    0    0]
 [    0    0   23    0    0]
 [    0    0    0    8    0]
 [    0    0    0    0    5]]
```

Predicted Values: -

Predicted values for Decision Tree:

[0 0 1 ... 0 0 0]

Predicted values for Random Forest:

[0 0 1 ... 0 0 0]

Predicted values for K-Nearest Neighbors:

[0 0 1 ... 0 0 0]

Predicted values for Logistic Regression:

[0 0 1 ... 0 0 0]

Predicted values for Support Vector Machine:

[0 0 1 ... 0 0 0]

Conclusion: -

In this predictive maintenance project for industrial equipment, we employed five different machine learning models: Decision Tree, Random Forest Classifier, k-Nearest Neighbors Classifier, Logistic Regression, and Support Vector Classifier. Remarkably, all models achieved perfect performance metrics, with an accuracy, precision, recall, and F1 score of 1.0.

This exceptional performance indicates that the models were highly effective in predicting equipment failures, ensuring timely maintenance interventions. The success of these models can be attributed to the high-quality data, effective feature engineering, and robust model training and validation processes.

Key Takeaways:

1. **Model Performance:** All five models demonstrated perfect predictive capabilities, highlighting their suitability for this application.
2. **Data Quality:** The high-quality and comprehensive dataset played a crucial role in achieving these results.
3. **Feature Engineering:** The careful selection and engineering of features ensured that the models could accurately capture the patterns indicative of equipment failures.
4. **Robust Validation:** Rigorous cross-validation techniques ensured that the models were not overfitting and could generalize well to new data.

Future Work: While the current models have shown excellent performance, continuous monitoring and updating of the models will be essential to maintain their effectiveness as new data becomes available. Additionally, exploring more advanced techniques and integrating real-time data processing could further enhance the predictive maintenance system.

Overall, this project demonstrates the potential of machine learning in transforming maintenance strategies, leading to reduced downtime, optimized maintenance schedules, and significant cost savings.